

1 Nonproprietary Names

BP: Ethanol (96%)

JP: Ethanol

PhEur: Ethanol (96 per cent)

USP: Alcohol

2 Synonyms

Ethanolum (96 per centum); ethyl alcohol; ethyl hydroxide; grain alcohol; methyl carbinol.

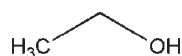
3 Chemical Name and CAS Registry Number

Ethanol [64-17-5]

4 Empirical Formula and Molecular Weight

C₂H₆O 46.07

5 Structural Formula



6 Functional Category

Antimicrobial preservative; disinfectant; skin penetrant; solvent.

7 Applications in Pharmaceutical Formulation or Technology

Ethanol and aqueous ethanol solutions of various concentrations (see Sections 8 and 17) are widely used in pharmaceutical formulations and cosmetics; see Table I. Although ethanol is primarily used as a solvent, it is also employed as a disinfectant, and in solutions as an antimicrobial preservative.^(1,2) Topical ethanol solutions are used in the development of transdermal drug delivery systems as penetration enhancers.⁽³⁻¹⁰⁾ Ethanol has also been used in the development of transdermal preparations as a co-surfactant.⁽¹¹⁻¹³⁾

Table I: Uses of alcohol.

Use	Concentration (% v/v)
Antimicrobial preservative	≥ 10
Disinfectant	60–90
Extracting solvent in galenical manufacture	Up to 85
Solvent in film coating	Variable
Solvent in injectable solutions	Variable
Solvent in oral liquids	Variable
Solvent in topical products	60–90

8 Description

In the BP 2009, the term ‘ethanol’ used without other qualification refers to ethanol containing ≥99.5% v/v of C₂H₆O. The term ‘alcohol’, without other qualification, refers to ethanol 95.1–96.9% v/v. Where other strengths are intended, the term ‘alcohol’ or ‘ethanol’ is used, followed by the statement of the strength.

In the PhEur 6.0, anhydrous ethanol contains not less than 99.5% v/v of C₂H₆O at 20°C. The term ethanol (96%) is used to describe the material containing water and 95.1–96.9% v/v of C₂H₆O at 20°C.

In the USP 32, the term ‘dehydrated alcohol’ refers to ethanol ≥99.5% v/v. The term ‘alcohol’ without other qualification refers to ethanol 94.9–96.0% v/v.

In the JP XV, ethanol (alcohol) contains 95.1–96.9% v/v (by specific gravity) of C₂H₆O at 15°C.

In the *Handbook of Pharmaceutical Excipients*, the term ‘alcohol’ is used for either ethanol 95% v/v or ethanol 96% v/v.

Alcohol is a clear, colorless, mobile, and volatile liquid with a slight, characteristic odor and burning taste.

See also Section 17.

9 Pharmacopeial Specifications

See Table II. See also Sections 17 and 18.

Table II: Pharmacopeial specifications for alcohol.

Test	JP XV	PhEur 6.0	USP 32
Identification	+	+	+
Characters	—	+	—
Specific gravity	0.809–0.816	0.805–0.812	0.812–0.816
Acidity or alkalinity	+	+	+
Clarity and color of solution	+	+	+
Nonvolatile residue	≤2.5 mg	≤25 ppm	≤2.5 mg
Volatile impurities	+	+	+
Absorbance	+	+	+
at 240 nm	≤0.40	≤0.40	≤0.40
at 250–260 nm	≤0.30	≤0.30	≤0.30
at 270–340 nm	≤0.10	≤0.10	≤0.10
Assay	95.1–96.9%	95.1–96.9%	92.3–93.8% by weight 94.9–96.0% by volume

10 Typical Properties

Antimicrobial activity Ethanol is bactericidal in aqueous mixtures at concentrations between 60% and 95% v/v; the optimum concentration is generally considered to be 70% v/v. Antimicrobial activity is enhanced in the presence of edetic acid or edetate salts.⁽¹⁾ Ethanol is inactivated in the presence of nonionic surfactants and is ineffective against bacterial spores.

Boiling point 78.15°C

Flammability Readily flammable, burning with a blue, smokeless flame.

Flash point 14°C (closed cup)

NIR spectra see Figures 1 and 2.

Solubility Miscible with chloroform, ether, glycerin, and water (with rise of temperature and contraction of volume).

Specific gravity 0.8119–0.8139 at 20°C

Note The above typical properties are for alcohol (ethanol 95% or 96% v/v). See Section 17 for typical properties of dehydrated alcohol.

11 Stability and Storage Conditions

Aqueous ethanol solutions may be sterilized by autoclaving or by filtration and should be stored in airtight containers, in a cool place.

12 Incompatibilities

In acidic conditions, ethanol solutions may react vigorously with oxidizing materials. Mixtures with alkali may darken in color owing to a reaction with residual amounts of aldehyde. Organic

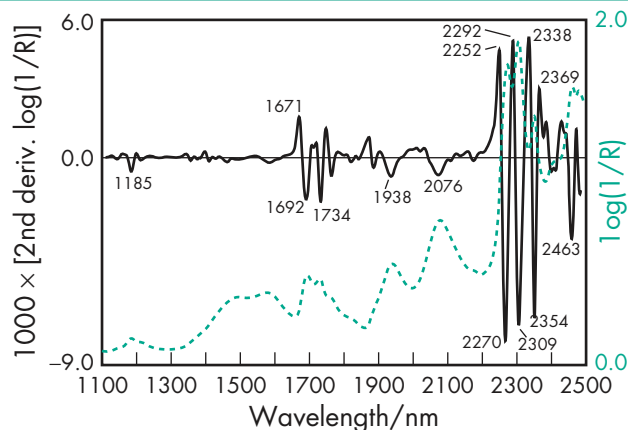


Figure 1: Near-infrared spectrum of alcohol (96%) measured by transfectance (1 mm path-length).

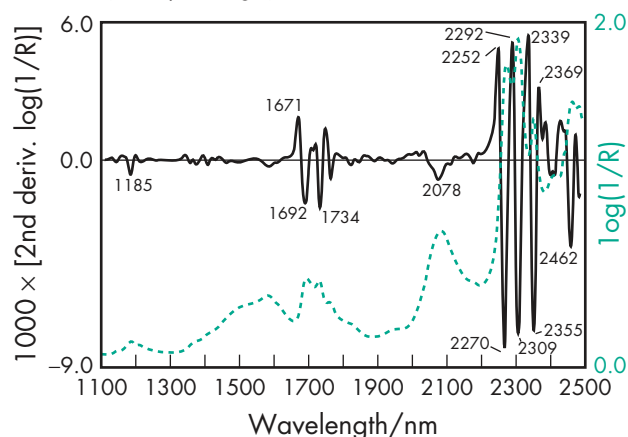


Figure 2: Near-infrared spectrum of alcohol (absolute) measured by transfectance (1 mm path-length).

salts or acacia may be precipitated from aqueous solutions or dispersions. Ethanol solutions are also incompatible with aluminum containers and may interact with some drugs.

13 Method of Manufacture

Ethanol is manufactured by the controlled enzymatic fermentation of starch, sugar, or other carbohydrates. A fermented liquid is produced containing about 15% ethanol; ethanol 95% v/v is then obtained by fractional distillation. Ethanol may also be prepared by a number of synthetic methods.

14 Safety

Ethanol and aqueous ethanol solutions are widely used in a variety of pharmaceutical formulations and cosmetics. It is also consumed in alcoholic beverages.

Ethanol is rapidly absorbed from the gastrointestinal tract and the vapor may be absorbed through the lungs; it is metabolized, mainly in the liver, to acetaldehyde, which is further oxidized to acetate.

Ethanol is a central nervous system depressant and ingestion of low to moderate quantities can lead to symptoms of intoxication including muscle incoordination, visual impairment, slurred speech, etc. Ingestion of higher concentrations may cause depression of medullary action, lethargy, amnesia, hypothermia, hypoglycemia, stupor, coma, respiratory depression, and cardiovascular collapse. The lethal human blood-alcohol concentration is generally estimated to be 400–500 mg/100 mL.

Although symptoms of ethanol intoxication are usually encountered following deliberate consumption of ethanol-containing beverages, many pharmaceutical products contain ethanol as a solvent, which, if ingested in sufficiently large quantities, may cause adverse symptoms of intoxication. In the USA, the maximum quantity of alcohol included in OTC medicines is 10% v/v for products labeled for use by people of 12 years of age and older, 5% v/v for products intended for use by children aged 6–12 years of age, and 0.5% v/v for products for use by children under 6 years of age.⁽¹⁴⁾

Parenteral products containing up to 50% of alcohol (ethanol 95 or 96% v/v) have been formulated. However, such concentrations can produce pain on intramuscular injection and lower concentrations such as 5–10% v/v are preferred. Subcutaneous injection of alcohol (ethanol 95% v/v) similarly causes considerable pain followed by anesthesia. If injections are made close to nerves, neuritis and nerve degeneration may occur. This effect is used therapeutically to cause anesthesia in cases of severe pain, although the practice of using alcohol in nerve blocks is controversial. Doses of 1 mL of absolute alcohol have been used for this purpose.⁽¹⁵⁾

Preparations containing more than 50% v/v alcohol may cause skin irritation when applied topically.

LD₅₀ (mouse, IP): 0.93 g/kg⁽¹⁶⁾

LD₅₀ (mouse, IV): 1.97 g/kg

LD₅₀ (mouse, oral): 3.45 g/kg

LD₅₀ (mouse, SC): 8.29 g/kg

LD₅₀ (rat, IP): 3.75 g/kg

LD₅₀ (rat, IV): 1.44 g/kg

LD₅₀ (rat, oral): 7.06 g/kg

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of material handled. Ethanol and aqueous ethanol solutions should be handled in a well-ventilated environment. In the UK, the long-term 8-hour TWA workplace exposure limit for ethanol is 1920 mg/m³ (1000 ppm).⁽¹⁷⁾ Ethanol may be irritant to the eyes and mucous membranes, and eye protection and gloves are recommended. Ethanol is flammable and should be heated with care. Fixed storage tanks should be electrically grounded to avoid ignition from electrostatic discharges when ethanol is transferred.

16 Regulatory Status

Included in the FDA Inactive Ingredients Database (dental preparations; inhalations; IM, IV, and SC injections; nasal and ophthalmic preparations; oral capsules, solutions, suspensions, syrups, and tablets; rectal, topical, and transdermal preparations). Included in the Canadian List of Acceptable Non-medicinal Ingredients. Included in nonparenteral and parenteral medicines licensed in the UK.

17 Related Substances

Dehydrated alcohol; denatured alcohol; dilute alcohol; isopropyl alcohol.

Dehydrated alcohol

Synonyms Absolute alcohol; anhydrous ethanol; ethanol.

Autoignition temperature 365°C

Boiling point 78.5°C

Explosive limits 3.5–19.0% v/v in air

Flash point 12°C (closed cup)

Melting point –112°C

Moisture content Absorbs water rapidly from the air.

Refractive index n_D^{20} = 1.361

Specific gravity 0.7904–0.7935 at 20°C

Surface tension 22.75 mN/m at 20°C (ethanol/vapor)

Vapor density (relative) 1.59 (air = 1)

Vapor pressure 5.8 Pa at 20°C

Viscosity (dynamic) 1.22 mPa s (1.22 cP) at 20°C

Comments Dehydrated alcohol is ethanol $\geq 99.5\%$ v/v. See Section 8. Dehydrated alcohol is one of the materials that have been selected for harmonization by the Pharmacopeial Discussion Group. For further information see the General Information Chapter <1196> in the USP32–NF27, the General Chapter 5.8 in PhEur 6.0, along with the ‘State of Work’ document on the PhEur EDQM website, and also the General Information Chapter 8 in the JP XV.

Denatured alcohol

Synonyms Industrial methylated spirit; surgical spirit.

Comments Denatured alcohol is alcohol intended for external use only. It has been rendered unfit for human consumption by the addition of a denaturing agent such as methanol or methyl isobutyl ketone.

Dilute alcohol

Synonyms Dilute ethanol.

Specific gravity see Table III.

Table III: Specific gravity of alcohol.

Strength of alcohol (% v/v)	Specific gravity at 20°C
90	0.8289–0.8319
80	0.8599–0.8621
70	0.8860–0.8883
60	0.9103–0.9114
50	0.9314–0.9326
45	0.9407–0.9417
25	0.9694–0.9703
20	0.9748–0.9759

Comments The term ‘dilute alcohol’ refers to a mixture of ethanol and water of stated concentration. The USP32–NF27 lists diluted alcohol. The BP 2009 lists eight strengths of dilute alcohol (dilute ethanol) containing 90%, 80%, 70%, 60%, 50%, 45%, 25%, and 20% v/v respectively of ethanol.

18 Comments

Alcohol is one of the materials that have been selected for harmonization by the Pharmacopeial Discussion Group. For further information see the General Information Chapter <1196> in the USP32–NF27, the General Chapter 5.8 in PhEur 6.0, along with the ‘State of Work’ document on the PhEur EDQM website, and also the General Information Chapter 8 in the JP XV.

Possession and use of nondenatured alcohols are usually subject to close control by excise authorities.

A specification for alcohol is contained in the Food Chemicals Codex (FCC).⁽¹⁸⁾

The EINECS number for alcohol is 200-578-6. The PubChem Compound ID (CID) for alcohol is 702.

19 Specific References

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20 General References

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21 Author

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22 Date of Revision

5 February 2009.



1 Nonproprietary Names

BP: Alginic Acid

PhEur: Alginic Acid

USP-NF: Alginic Acid

2 Synonyms

Acidum alginicum; E400; *Kelacid*; L-gulo-D-mannoglycuronan; polymannuronic acid; *Protacid*; *Satialgine H8*.

3 Chemical Name and CAS Registry Number

Alginic acid [9005-32-7]

4 Empirical Formula and Molecular Weight

Alginic acid is a linear glycuronan polymer consisting of a mixture of β -(1 $\rightarrow$ 4)-D-mannosyluronic acid and α -(1 $\rightarrow$ 4)-L-gulosyluronic acid residues, of general formula $(C_6H_8O_6)_n$. The molecular weight is typically 20 000–240 000.

5 Structural Formula

The PhEur 6.3 describes alginic acid as a mixture of polyuronic acids $[(C_6H_8O_6)_n]$ composed of residues of D-mannuronic and L-glucuronic acid, and obtained mainly from algae belonging to the Phaeophyceae. A small proportion of the carboxyl groups may be neutralized.

See also Section 4.

6 Functional Category

Release-modifying agent; stabilizing agent; suspending agent; sustained release agent; tablet binder; tablet disintegrant; taste-masking agent; viscosity-increasing agent.

7 Applications in Pharmaceutical Formulation or Technology

Alginic acid is used in a variety of oral and topical pharmaceutical formulations. In tablet and capsule formulations, alginic acid is used as both a binder and disintegrating agent at concentrations of 1–5% w/w.^(1,2) Alginic acid is widely used as a thickening and suspending agent in a variety of pastes, creams, and gels; and as a stabilizing agent for oil-in-water emulsions.

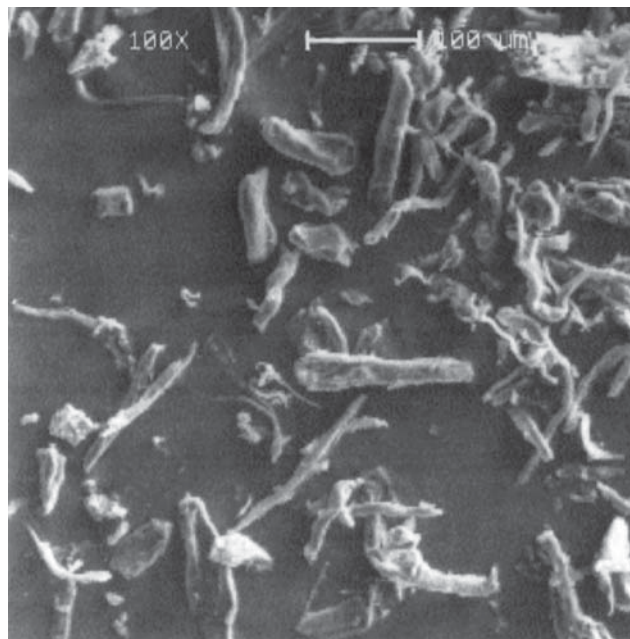
Alginic acid has been used to improve the stability of levosimendan.⁽³⁾

Therapeutically, alginic acid has been used as an antacid.⁽⁴⁾ In combination with an H_2 -receptor antagonist, it has also been utilized for the management of gastroesophageal reflux.⁽⁵⁾

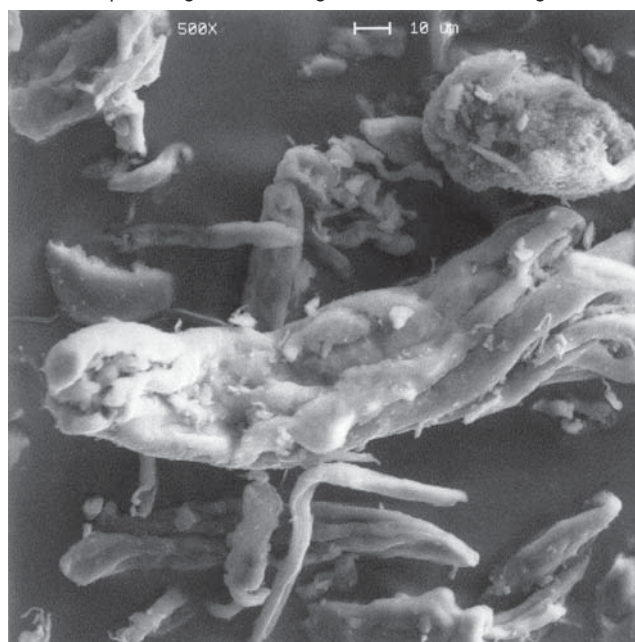
8 Description

Alginic acid is a tasteless, practically odorless, white to yellowish-white, fibrous powder.

SEM 1: Excipient: alginic acid; magnification: 100 $\times$; voltage: 25 kV.



SEM 2: Excipient: alginic acid; magnification: 500 $\times$; voltage: 25 kV.



9 Pharmacopeial Specifications

See Table I.

10 Typical Properties

Acidity/alkalinity pH = 1.5–3.5 for a 3% w/v aqueous dispersion.

Crosslinking Addition of a calcium salt, such as calcium citrate or calcium chloride, causes crosslinking of the alginic acid polymer